# An Information-Theoretic Latent Disentanglement Framework for Multimodal Hyperspectral Image Compression

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***Abstract*—Existing single-modality hyperspectral image compression methods often lack explicit structural constraints in the latent space. Most methods rely on pixel-level rate–distortion objectives. Spatial–spectral consistency with physical meaning becomes difficult to preserve under high compression ratios. Multimodal representation learning often depends on empirical modeling. Theoretical guidance remains limited. To address these issues, an information-theoretic constrained framework for multimodal hyperspectral image compression is proposed. Auxiliary modality information provides structural guidance for the latent space. Latent representations are explicitly disentangled into modality-specific and shared components. Mutual information-based regularization is introduced to suppress statistical dependence. The regularization is integrated into the rate–distortion optimization framework. In addition, a best neighbor matching mechanism is introduced. Cross-modal spatial features are aligned. Structural consistency in the latent space is further enhanced. Experiments on mixed multimodal hyperspectral datasets demonstrate superior performance over existing single-modality compression methods. Improvements are observed in both rate–distortion performance and spectral consistency.**

***Index Terms*—Hyperspectral image, image compression, latent disentanglement, latent-space regularization, multimodal learning, mutual information constraint.**

## I. Introduction

HYPERSPECTRAL imagery (HSI) is often referred to as an image cube, consisting of spatial and spectral dimensions. Higher spectral resolution provides detailed spectral features. These enable effective target identification and discrimination. The versatility of HSI enables its broad application in land classification [1], environmental monitoring [2], and disaster warning [3]. With the growing availability of spaceborne observation platforms, hyperspectral technology has become an important tool for Earth observation.

Satellite platforms, however, face constraints in sensor payload, computational resources, and communication bandwidth. Hyperspectral data inevitably undergoes compression during transmission, which leads to information loss. This compromises the data's integrity and reliability, which in turn hinders its effectiveness in subsequent applications. The inherent spectral redundancy and high dimensionality of hyperspectral data complicate the compression process. Traditional methods often fail to retain key information, causing further information loss and reduced reconstruction performance.

At low bit rates, traditional satellite-oriented compression algorithms (e.g., JPEG2000 [4], JPEG-LS [5], and BPG [6]) fail to preserve spatial continuity and spectral consistency. This limitation leads to severe degradation in perceptual quality. These distortions reduce the perceptual quality of the reconstructed images and compromise the reliability of downstream tasks, including target recognition [7] and classification [8].

Recent advances in deep learning have further promoted the evolution of HSI compression methods [9]. They have attracted growing scholarly attention and become a prominent research focus. Existing single-modality HSI compression methods [10], [11] largely follow the modeling paradigm of natural image compression. Deep neural networks are employed to learn compact latent representations. Entropy models are subsequently applied for coding. Several studies introduce improvements tailored to hyperspectral data characteristics, including spatial–spectral joint modeling [10], [12] and nonlocal similarity exploitation [11]. These efforts focus on network architecture design or feature enhancement. Some studies attempt to improve entropy models for hyperspectral compression [13], but these approaches introduce considerable computational overhead.

Under single-modality compression frameworks, latent variables exhibit complex and highly coupled statistical dependencies. The paradigm remains a black-box representation learning framework with latent representations modeled as a whole, without explicit characterization or constraint of internal statistical structures. The paradigm also suffers from limited structural awareness. Most models rely primarily on spectral statistics and lack external spatial or geometric references. This limitation results in ambiguous latent representations and insufficient preservation of the true spatial structure of the scene.

In contrast, multimodal compression mitigates the limitations of single-modality frameworks. Complementary modalities, including multispectral imagery (MSI), synthetic aperture radar (SAR), and light detection and ranging (LiDAR),

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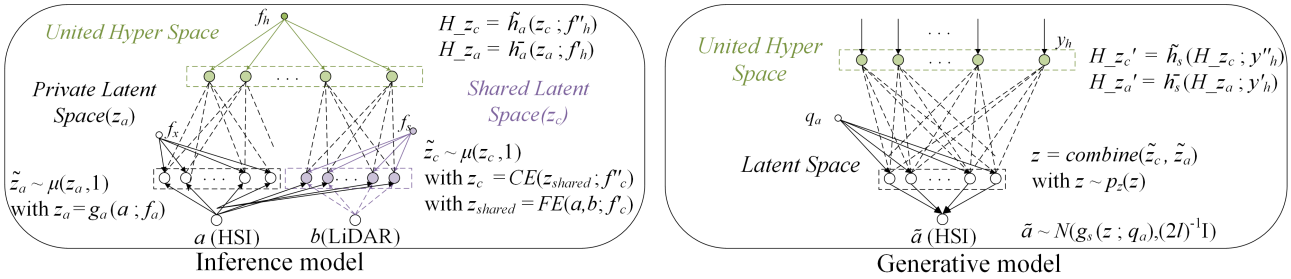



Fig. 1. Diagram of the proposed multimodal disentangled compression framework. Features are divided into two groups: domain-specific representations ($z_a$) and shared representations ($z_c$). Importantly, the auxiliary modality (LiDAR) is employed solely to derive shared representations and enhance the encoding of the latent space, without affecting the compression process of the primary modality.

provide spatial, spectral, and structural cues that enhance the integrity of compressed representations. LiDAR-derived elevation information reduces spectral ambiguity in hyperspectral data and enables discrimination between objects with similar spectral signatures but different geometric structures [19]. Studies have shown that multimodal fusion improves remote sensing performance. Ma et al. [20] achieved higher segmentation accuracy with a hierarchical Transformer, while Roy et al. [21] enhanced classification generalization using LiDAR-assisted Vision Transformers. Fusing heterogeneous modalities enables more comprehensive feature learning and leads to performance gains in complex tasks. Existing multimodal representation learning methods, including fusion-based approaches [14], [15], disentanglement methods [16], and related frameworks [17], achieve strong performance in classification and understanding tasks. These methods rely on data-driven feature learning, and some approaches separate shared and modality-specific information [18], which improves cross-modal representation.

However, existing multimodal methods primarily target high-level semantic tasks, and compression objectives are not considered. Rate–distortion optimization is not explicitly modeled, and redundant information across modalities is often encoded repeatedly. In multimodal compression, shared and modality-specific information is highly coupled in the latent space. This coupling results in insufficiently structured representations and introduces ambiguity in latent variable selection. Training becomes unstable, and compression efficiency is reduced. Existing entropy models cannot distinguish information sources across modalities and only estimate the distribution of latent variables. The relationship between latent statistical structure and coding efficiency remains unclear. As a result, direct application to multimodal image compression remains challenging.

To address these issues, the proposed method focuses on latent-space modeling and optimization objectives rather than network architecture design, which is shown in Fig. 1. Information-theoretic constraints are introduced to explicitly characterize the statistical structure of latent representations. A unified modeling framework is constructed for shared and modality-specific information based on disentangled representations.

The optimization objective is derived from information theory and integrated into the rate–distortion framework as regularization terms. These constraints enforce structured latent representations and reduce statistical mismatch with entropy model assumptions. Existing entropy models can be directly applied for entropy estimation. Redundant encoding is suppressed and bit allocation is simplified. This design achieves joint optimization of compression efficiency and information preservation. The shared-private representation paradigm is extended from multimodal learning to compression. A theoretically grounded and interpretable rate–distortion modeling framework is established.

To the best of current knowledge, we present the first framework for multimodal hyperspectral image compression. It introduces information-theoretic disentangled latent modeling into compression and provides a unified approach with both interpretability and practical effectiveness.

The main contributions of this work can be summarized as follows.

1) A multimodal disentangled compression framework for hyperspectral images is introduced with LiDAR as an auxiliary modality. The model captures shared and modality-specific features in the latent space. This design reduces redundancy and improves cross-modal disentanglement. It establishes a principled framework for multimodal latent representation learning.
2) A theoretical derivation and optimization framework is developed based on maximum marginal likelihood. The variational lower bound is formulated under a minimum mutual information constraint. The resulting objective function is interpretable and guides latent-space disentanglement and model design.
3) An optimal nearest-neighbor matching mechanism is designed to refine shared representations. The model performs dynamic neighborhood search and selects top-k auxiliary blocks based on Euclidean distance. This mechanism enhances local structure alignment and improves cross-modal shared feature modeling.

## II. Related Work

### A. Multimodal Disentanglement Techniques

Recent studies demonstrate the effectiveness of multimodal disentanglement frameworks in multichannel emotion recognition (MMER) and multimodal sentiment analysis (MSA). MMER integrates information from multiple perceptual modalities to improve emotion recognition accuracy, and MSA exploits complementary features from speech, text,

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and video to support comprehensive emotional modeling and deeper understanding of user-generated content. However, multimodal data exhibit inherent heterogeneity and disparity, which leads to inconsistent and imbalanced representations.

Zhou et al. [23] proposed TriDiRA, which decomposes features into invariant, effective, and ineffective modality-specific components and fuses the invariant and effective parts to suppress redundant or inconsistent information. Sun et al. [24] introduced a distribution alignment strategy that maps representations from different modalities into independent subspaces and incorporates cross-modal fusion with multiple regularization constraints to improve stability and robustness. Ding et al. [25] proposed a modality-invariant progressive representation framework, which decouples multimodal features into a frequency band-based invariant domain and enables high-precision image registration through multiscale field estimation.

Despite significant progress in modal disentanglement, existing methods still exhibit several limitations. Many approaches cannot effectively handle redundant or conflicting information within individual modalities. This issue disrupts the structural organization of the latent space and reduces representational capacity in complex scenarios. In addition, many methods rely on fixed feature extraction architectures and emphasize loss function design. Most studies provide loss formulations without detailed mathematical derivations. This limitation increases implementation complexity and restricts generalizability across domains.

### B. Learning-Based HSI Compression Method

In recent years, the field of hyperspectral imaging has achieved rapid progress, driven by advances in computational imaging and data-driven methodologies. Hong et al. [26] provide a comprehensive review of hyperspectral imaging from the perspectives of computational imaging, physics-informed modeling, and cross-modal fusion. It offers new insights into the acquisition and modeling of multimodal hyperspectral data. Multimodal learning techniques have since emerged as a major research focus. Lin et al. [27] introduce the mixture-of-experts framework into multimodal remote sensing. Task decomposition and expert collaboration address the complexity and diversity of visual and textual modalities. Zhou et al. [15] propose a collaborative high-order interaction paradigm. This approach models multimodal relationships at both fine-grained spatial and global statistical levels. These studies advance multimodal representation learning from multiple perspectives. They also provide a solid theoretical foundation for information modeling in complex scenarios.

In this context, studies on other hyperspectral-related tasks offer useful insights for compression. SpectralGPT [28] introduces a 3-D token generation mechanism for joint spatial–spectral modeling. A multiobjective reconstruction strategy is employed to build a remote sensing foundation model. LRR-Net [29] adopts an interpretable deep unfolding framework for hyperspectral anomaly detection. This design improves performance and enhances model interpretability. These works highlight the importance of joint spatial–spectral modeling and structured representation learning in hyperspectral data processing. They also demonstrate the value of interpretability in complex model design.

The above methods have been applied to hyperspectral images in tasks such as classification, reconstruction, and anomaly detection. They provide useful insights for HSI compression. Existing HSI compression approaches can be broadly divided into four categories.

*1) Preservation of Structural Details and Spatial Information:* Recent studies integrate spatial feature enhancement mechanisms to preserve geometric structures and texture characteristics while reducing bitrate. Dua et al. [30] and La Grassa et al. [31] proposed convolutional autoencoder-based HSI compression frameworks, which improve visual quality and perceptual accuracy over traditional methods. Guo et al. [32] proposed an edge-guided compression model with edge information regularization. The model emphasizes critical texture regions and improves detail preservation.

*2) Balancing Spectral and Spatial Fidelity:* HSIs contain highly correlated spectral features and rich spatial structures. Joint spectral–spatial modeling has been widely studied to balance spectral consistency, visual quality, and compression efficiency. Guo et al. [33] developed a contrastive learning-based self-supervised framework with improved spectral consistency. Liu et al. [9] proposed a generative dual-residual compression model. The model produces compact latent representations that encode band priors and spatial structures. These representations serve as conditional guidance in diffusion-based reconstruction and support joint preservation of spectral and spatial information.

*3) Lightweight Design and Satellite Compatibility:* Recent work focuses on lightweight models for deployment under constrained computational and storage resources. Mijares i Verdú et al. [34] proposed a spectral grouping strategy that partitions bands into compact groups for network input. The method improves parameter efficiency and maintains reconstruction accuracy. Liu et al. [35] developed a recurrent compression model for band-wise processing. The design reduces parameters and supports efficient lightweight implementation.

*4) Explicit and Implicit Neural Representations:* Recent studies incorporate explicit and implicit neural representations into HSI compression. Zhang et al. [36] proposed FHN-erf, which maps spatial coordinates to pixel values using a multilayer perceptron. The model learns coordinate-to-signal representations and reduces parameter complexity. Zhao et al. [37] modeled spectral bands as sequential image frames and performed compression through sequence-level modeling.

Prompted by these advances, HSI compression has developed rapidly. Building on this progress, a multimodal disentangled compression model is introduced with HSIs as the primary modality and LiDAR data as the auxiliary modality.

## III. Theoretical Foundations

This section presents the proposed multimodal disentanglement compression model based on latent-space regularization and its implementation framework. The limitations of unified latent space in multimodal compression are first analyzed, including cross-modal redundancy, representation strategy, and

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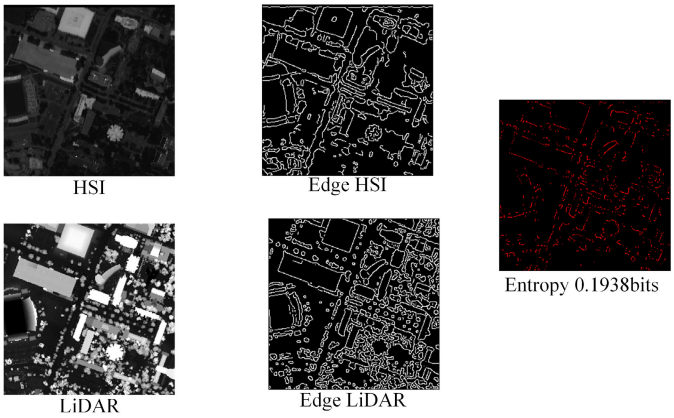



Fig. 2. Quantitative analysis of modal redundancy.

information preservation. A solution based on maximum marginal likelihood optimization is then introduced.

A regularization constraint is imposed on the latent space from the perspective of minimal mutual information. This constraint leads to the derivation of an objective function for the multimodal compression framework. Finally, an optimal neighbor matching mechanism is designed to refine shared representations.

### A. Motivation: Limitations of Unified Latent Space in Multimodal Compression

Learned compression models are predominantly developed under a single-modality assumption. Extending these models to multimodal scenarios often leads to a unified latent representation design. This formulation introduces several intrinsic limitations. In particular, it does not explicitly account for cross-modal redundancy, provides limited flexibility for heterogeneous representation learning, and may degrade information preservation during compression. We analyze these limitations from three perspectives: cross-modal redundancy, representation strategy, and information preservation.

*1) Cross-Modal Redundancy:* In practical remote sensing scenarios, HSI, multispectral images, and SAR data are often acquired in a coordinated manner. These modalities exhibit strong correlations in both spatial structure and semantic content. Existing hyperspectral image compression methods, however, remain largely restricted to single-modality modeling. Such approaches fail to exploit complementary information across modalities. When extended to multimodal compression, cross-modal redundancy becomes a primary challenge. For instance, edge and structural features in SAR images are also present in HSI. Independent compression of each modality, or indiscriminate joint compression, leads to repeated encoding of shared information. This redundancy results in bitrate inefficiency and degraded compression performance.

As illustrated in Fig. 2, cross-modal redundancy between HSI and the corresponding LiDAR modality is analyzed from the perspective of high-frequency information. Edge features are extracted from both modalities to represent high-frequency structural components. The similarity and entropy distributions are then computed to quantify the degree of information

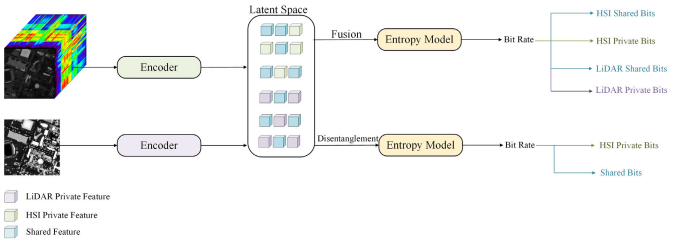



Fig. 3. Bitrate required by different encoding strategies.

overlap. High-frequency information typically corresponds to edges, textures, and structural discontinuities. These components reflect the geometric contours and spatial structures of ground objects.

In remote sensing scenarios, both HSI and LiDAR describe the same physical scene. This shared physical basis leads to strong correlations in high-frequency structures. For example, building boundaries, terrain variations, and vegetation distributions appear as edge or gradient variations in both modalities.

Experimental results demonstrate strong consistency between HSI and LiDAR in edge regions. Their high-frequency components exhibit significant statistical correlation.

*2) Representation Disentanglement:* Multimodal data exhibit both substantial inter-modal correlations and modality-specific information components. In the absence of an effective disentanglement mechanism, a clear distinction between cross-modal shared information and modality-specific information remains difficult. This entanglement leads to redundant encoding. It also disrupts accurate entropy modeling of latent distributions. These effects limit further improvement in compression performance. Effective disentanglement of latent representations serves as a key prerequisite for efficient multimodal compression.

Existing multimodal approaches, including those designed for multimodal classification, commonly adopt feature fusion strategies. These methods enhance representation capability through joint modeling of multisource information. However, they rely on implicit feature learning and lack explicit structural modeling and constraints on latent representations. In multimodal compression scenarios, such fusion strategies promote further coupling between shared and modality-specific information in the latent space. This effect weakens redundancy suppression and reduces the precision of bit allocation. A latent representation model with explicit structural constraints is required. This design enables effective separation between shared and private information and improves overall compression efficiency.

From an information-theoretic perspective, compression performance can be characterized by the coding entropy of latent representations. In fusion-based strategies, the latent representation is typically formulated as $z = f(x_a, x_b)$, where $x_a$ denotes the primary modality (HSI) and $x_b$ denotes the auxiliary modality (LiDAR). The absence of explicit structural constraints results in latent representations that contain both shared and modality-specific information. Redundant components are inevitable. Fig. 3 illustrates the theoretical difference in required coding representations between the two strategies.

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The bitrate of the fusion strategy can be approximated as

$$R_{fusion} \approx H(z) = H(z_c) + H(z_a) + H(z_b) + H(z_r) \quad (1)$$

where $z_c$ denotes cross-modal shared information, $z_a$ and $z_b$ denote modality-specific information, and $z_r$ denotes redundant information induced by entanglement.

In contrast, the disentanglement strategy constructs an explicit latent variable decomposition $z = (z_c, z_a, z_b)$. It introduces independence constraints, such as $z_a \perp z_c, z_b \perp z_c$. This design ensures single encoding of shared information through $z_c$. In compression tasks with HSI as the primary modality, auxiliary modality information does not participate in encoding. This mechanism avoids additional bitrate contributions. The resulting bitrate can be approximated as

$$R_{dis} \approx H(z_c) + H(z_a). \quad (2)$$

From the perspective of coding structure, the fusion strategy encodes shared information, modality-specific information from both modalities, and redundant components introduced by entanglement. The disentanglement strategy encodes only shared information and modality-specific information of the primary modality. This structure yields a more compact latent representation.

The bitrate difference between the two strategies can be expressed as

$$\Delta R = R_{\text{fusion}} - R_{\text{dis}} \geq 0. \quad (3)$$

The non-negativity of entropy indicates additional coding cost introduced by the redundant term $z_r$ in the fusion strategy. The disentanglement strategy removes this redundant component through explicit separation of shared and modality-specific information. This property leads to improved compression efficiency from an information-theoretic perspective.

Motivated by these observations, structured constraints on latent variables are introduced. The proposed framework incorporates a mutual information-based disentanglement assumption together with a shared-private separation principle. Under these constraints, latent variables follow a structured information allocation scheme. Cross-modal shared information is represented exclusively by the shared variable $z_c$. Modality-specific information is encoded by $z_a$ and $z_b$. This structural constraint enforces statistical decomposability of latent representations. It provides a necessary condition for entropy minimization during encoding.

*3) Information Preservation:* Under high compression ratios, the reconstruction quality of HSI faces significant challenges. HSI exhibits highly continuous spectral structures, with strong correlations between adjacent bands. Insufficient modeling of spectral information during compression can lead to discontinuities or distortions in spectral curves, which in turn reduces the reliability of downstream tasks such as classification and detection.

In multimodal scenarios, where HSI is combined with other modalities such as LiDAR, additional challenges arise. Each modality contains modality-specific information that may be over-smoothed or partially lost during compression. As a result, the reconstructed multimodal data may suffer from reduced class separability and weakened spectral–physical consistency across modalities.

### B. Multimodal Disentangled Compression Framework

Building upon the above analysis, disentangled representation learning is introduced into multimodal hyperspectral image compression. The compression task is formulated as a latent variable estimation problem under a maximum marginal likelihood principle. The overall framework is illustrated in Fig. 4.

In the multimodal compression model, $g_a$ encodes HSI into latent features $z_a$ representing domain-specific information. A shared feature extractor (FE) processes HSI and LiDAR inputs and produces shared representations ($F_a$ and $F_b$). Private encoders (PEs) and a shared encoder common_encoder (CE) map these features to latent variables, including domain-specific components ($z_{c_a}$ and $z_{c_b}$) and shared variables ($z_c$).

A Kullback–Leibler (KL)-based regularization aligns $z_{c_a}$, $z_{c_b}$, and $z_c$ in the latent space. The priors $p(z_a)$ and $p(z_c)$ are estimated by $h_a$ and $h_s$ for bitrate computation. The decoder $g_s$ reconstructs the image from $z_a$ and $z_c$. LiDAR data serve as auxiliary supervision and enforce latent-space consistency.

The classic variational image compression framework (see [44]) is adopted as the baseline. A multimodal disentanglement mechanism and the corresponding regularization loss are introduced. The remaining network structure remains consistent with the baseline. The theoretical derivation of this architecture is presented in the following.

### C. Theoretical Derivation

In the proposed multimodal compression framework, observations are defined as a pair $(a, b)$, where $a$ denotes the primary modality and $b$ denotes the auxiliary modality. In this study, $a$ corresponds to hyperspectral images and $b$ corresponds to LiDAR data.

The compression model is optimized by maximizing the joint marginal likelihood of $(a, b)$, which aligns the learned distribution with the multimodal data distribution. The observations originate from two domains, denoted as $A$ and $B$.

Latent variables are disentangled to capture distinct information. Statistical independence is imposed among latent variables. This assumption is reflected in the conditional decomposition of the generative model. The decoder models latent variables through independent conditional distributions, expressed as follows:

$$p(a, b) = p(a \mid z_a, z_c)\ p(b \mid z_b, z_c)\ p(z_a)\ p(z_c)\ p(z_b) \quad (4)$$

where $p(a \mid z_a, z_c)$ and $p(b \mid z_b, z_c)$ denote the decoder generative models based on the latent variables $z_a$ and $z_b$.

Parameter learning is formulated as maximizing the joint marginal likelihood $p(a, b)$ of multimodal observations. This objective enforces statistical consistency between reconstructed data and observed samples.

The marginal probability $p(a, b)$ is obtained by integrating over latent variables $z = [z_a, z_c, z_b]$. The conditional likelihoods $p(a \mid z_a, z_c)$ and $p(b \mid z_b, z_c)$ describe the generative processes for domains $A$ and $B$. In the variational autoencoder (VAE) framework, these likelihoods are implemented by the decoder. The priors $p(z_a)$, $p(z_c)$, and $p(z_b)$ define the latent-space

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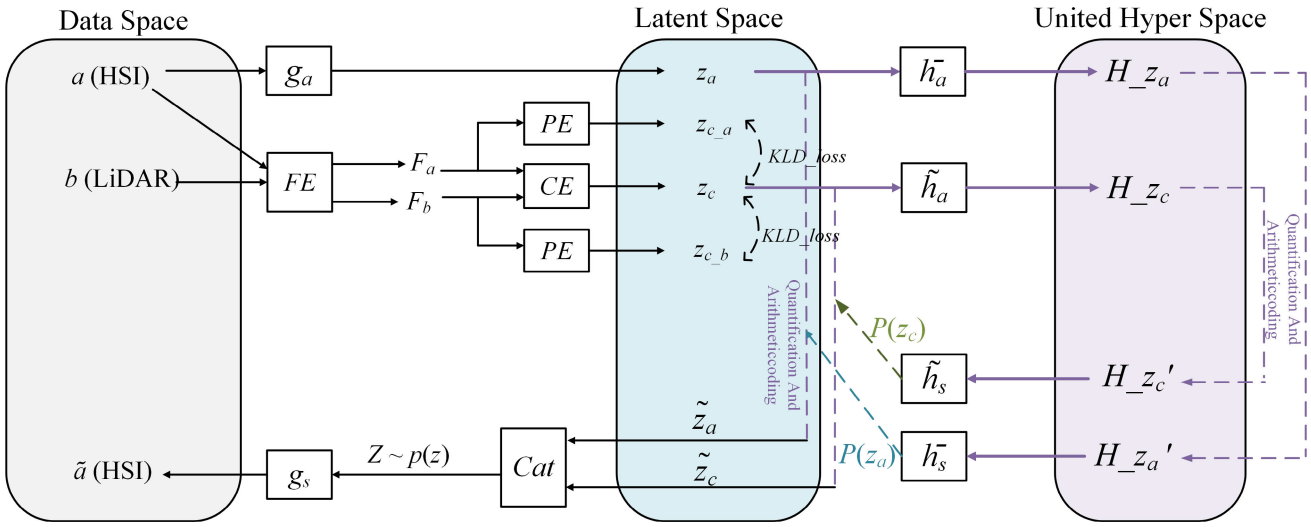



Fig. 4. Multimodal compression pipeline.

distribution. These priors are estimated through the hyperprior network and are modeled as Gaussian distributions [39].

The conditional distributions are parameterized by neural networks. The networks generate samples from latent variables $z_a$, $z_c$, and $z_b$ to approximate the data distribution.

Direct maximization of $p(a, b)$ is intractable. The computation requires integration over high-dimensional latent variables. The integrand is highly nonlinear due to neural network parameterization. Exact gradients cannot be obtained.

To address these challenges, a variational inference strategy is adopted [40]. An approximate posterior is introduced into the marginal likelihood. Jensen's inequality is applied to derive a tractable objective. The joint probability is factorized according to the dependencies among latent variables and observations. The factorized form is substituted into (5). This formulation yields the evidence lower bound (ELBO) as the optimization objective, expressed as

$$\begin{aligned}\mathrm{ELBO}(\theta,\phi) = {} & E\left[\log p\left(a \mid z_a, z_c\right)\right] + E\left[\log p\left(b \mid z_b, z_c\right)\right] \\ & - D_{KL}\left[q\left(z_a \mid a\right) \| \, p\left(z_a\right)\right] \\ & - D_{KL}\left[q\left(z_b \mid b\right) \| \, p\left(z_b\right)\right] \\ & - D_{KL}\left[q\left(z_c \mid a, b\right) \| \, p\left(z_c\right)\right]. \end{aligned} \tag{5}$$

During ELBO maximization, optimization focuses on learning latent variable distributions. The encoder distributions $q(z_a \mid a)$, $q(z_b \mid b)$, and $q(z_c \mid a, b)$ approximate the posterior through neural networks. These networks model latent distributions without explicit structural constraints.

The generative distribution in the latent space remains implicitly organized. Disentanglement among $z_a$, $z_c$, and $z_b$ is not explicitly enforced. ELBO maximization alone does not guarantee structured latent representations.

The KL divergence terms penalize the deviation between encoder posteriors and priors, including $q(z_a \mid a)$ and $p(z_a)$. This constraint promotes alignment with the prior distribution. It does not reduce dependencies among latent variables. It does not enforce disentanglement.

Effective disentanglement requires explicit latent-space design. Each latent variable should capture independent information, including domain-specific and shared components. The standard ELBO framework does not impose this requirement. The latent space may contain mixed and dependent representations. This issue motivates the following assumptions.

1) Mutual Disentanglement of $z_a$, $z_b$, and $z_c$: The generative factors captured by $z_a$, $z_b$, and $z_c$ should be independent to avoid redundant encoding.
2) Separation of $z_a$ and $z_c$: The latent variables $z_a$ and $z_b$ are intended to encapsulate generative factors that are specific to each respective domain, whereas the variable $z_c$ is designed to represent factors that are shared across domains.

To guide latent representation learning, information-theoretic regularization constraints are imposed on the approximate posterior $q$-distribution. Two constraints are defined.

The first constraint enforces separation between shared and domain-specific representations. Each latent variable encodes independent factors. Shared factors correspond to common information across modalities, and domain-specific factors correspond to modality-dependent information. This design reduces redundancy and prevents repeated encoding.

The constraint is implemented by minimizing mutual information between latent variables, including $I(z_a; z_c)$ and $I(z_b; z_c)$. These terms enforce statistical independence between shared and domain-specific representations.

From an information-theoretic perspective and considering the dependency of latent variables on the observed domain $a$, the joint mutual information can be decomposed as

$$I\left(z_a; z_c\right) = I\left(a; z_a\right) + I\left(a; z_c\right) - I\left(a; z_a, z_c\right) + I\left(z_a; z_c \mid a\right) \tag{6}$$

where $I(z_a; z_c \mid a)$ denotes the conditional mutual information.

Following the disentanglement requirement, the private and shared representations are assumed independent given $a$, yielding:

$$q\left(z_a, z_c \mid a\right) = q\left(z_a \mid a\right) \, q\left(z_c\right) \tag{7}$$

which implies $I(z_a; z_c \mid a) \approx 0$.

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Consequently, the mutual information reduces to a sum of domain-dependent and regularization terms

$$I(z_a; z_c) = -I(a; z_a, z_c) + I(a; z_a) + I(a; z_c). \tag{8}$$

Minimizing $I(z_a; z_c)$ reduces redundant information between private and shared latent variables and promotes disentanglement. This constraint does not guarantee effective separation in domain $A$. Degenerate solutions may occur. The latent variables may collapse and fail to encode meaningful information. Additional constraints are required to prevent this issue. Interaction information, also known as co-information [41], is introduced into the objective function. This measure captures higher order dependencies among latent variables.

Based on the definitions of mutual information and conditional mutual information, the formulation is expressed as

$$\begin{aligned} I(a; b; z_c) &= I(a; b) - I(a; b \mid z_c) \\ &= [H(a) - H(a \mid b)] - [H(a \mid z_c) - H(a \mid b, z_c)] \\ &= [H(a) - H(a \mid z_c)] - [H(a \mid b) - H(a \mid b, z_c)] \\ &= I(a; z_c) - I(a; z_c \mid b). \end{aligned} \tag{9}$$

This formulation illustrates that maximizing the interaction information encourages $z_c$ to encode common factors between modalities. Specifically, when $z_c$ captures informative content from $a$, the first term rises, while it decreases if similar information is already contained in $b$. The combined effect encourages $z_c$ to preserve only the information mutually shared between the two modalities.

The joint optimization of the two regularization terms can be expressed as

$$\begin{aligned} &\max\ I(a; b; z_c) - I(z_a; z_c) \\ &\quad = I(a; z_a, z_c) - I(a; z_a) - I(a; z_c \mid b). \end{aligned} \tag{10}$$

Direct optimization of this objective is challenging. The difficulty lies in the computation of mutual information terms, which depend on unknown data distributions and their marginals. For example, the first mutual information term can be written as

$$I(a; z_a, z_c) = D_{\mathrm{KL}}(p(a, z_a, z_c) \parallel p(a)\, p(z_a, z_c)). \tag{11}$$

Estimating the joint distribution $p(a, z_a, z_c)$ and its marginals $p(a)$ and $p(z_a, z_c)$ requires integration over the empirical data distribution $p(a, b)$. These distributions do not have analytical forms, making the integrations intractable.

The second term depends on $p(a)$ and $p(z)$, which are unknown. The third term involves the conditional distribution $p(a \mid b)$, which is also unavailable. Direct optimization of these mutual information terms is therefore infeasible. To address this, the optimization is reformulated using generative distributions and their variational lower bounds

$$\begin{aligned} I(a; z_a, z_c) &= \mathrm{E}_{p(a, z_a, z_c)}\left[\log \frac{p(a, z_a, z_c)}{p(a)\ p(z_a, z_c)}\right] \\ &= \mathrm{E}_{p(a, z_a, z_c)}\left[\log \frac{p(a \mid z_a, z_c)}{p(a)}\right]. \end{aligned} \tag{12}$$

Because the true joint distribution $p(a, z_a, z_c)$ is inaccessible, it is estimated through the empirical data distribution $p(a)$ combined with the inference distribution $q(z_a, z_c \mid a)$

$$p(a, z_a, z_c) \approx p(a)\ q(z_a, z_c \mid a). \tag{13}$$

Thus, the first term is approximated as

$$\begin{aligned} I(a; z_a, z_c) &\approx \mathrm{E}\left[\log \frac{q(a \mid z_a, z_c)}{p(a)}\right] \\ &= \mathrm{E}_{q(z_a, z_c \mid a)p(a)}\left[\log q(a \mid z_a, z_c)\right] \\ &\quad - \mathrm{E}_{q(z_a, z_c \mid a)p(a)}\left[\log p(a)\right] \\ &= H(a) + \mathrm{E}_{q(z_a, z_c \mid a)p(a)}\left[\log q(a \mid z_a, z_c)\right]. \end{aligned} \tag{14}$$

The second term in (10) is represented through the aggregated posterior

$$\begin{aligned} I_q(a; z_a) &= \mathrm{E}_{q(a, z_a)}\left[\log \frac{q(a, z_a)}{q(a)\, q(z_a)}\right] \\ &= \mathrm{E}_{p(a)}\left[D_{\mathrm{KL}}(q(z_a \mid a) \parallel q(z_a))\right]. \end{aligned} \tag{15}$$

During compression, the distribution $p(z_a)$ is modeled by the hyperprior and can be treated as known. The KL divergence with respect to this prior is expressed as

$$\begin{aligned} &\mathrm{E}_{p(a)}\left[D_{\mathrm{KL}}(q(z_a \mid a) \parallel p(z_a))\right] \\ &= \mathrm{E}_{p(a)}\mathrm{E}_{q(z_a \mid a)}\left[\log \frac{q(z_a \mid a)}{q(z_a)}\right] + \mathrm{E}_{q(z_a)}\left[\log \frac{q(z_a)}{p(z_a)}\right] \\ &= I_q(a; z_a) + D_{\mathrm{KL}}(q(z_a) \parallel p(z_a)). \end{aligned} \tag{16}$$

Since the KL divergence is non-negative, an upper bound for $I_q(a; z_a)$ is given as

$$-I_q(a; z_a) \geq -\mathrm{E}_{p(a)}\left[D_{\mathrm{KL}}(q(z_a \mid a) \parallel p(z_a))\right]. \tag{17}$$

The third term in (10) is also represented by the aggregated posterior

$$-I(a; z_c \mid b) = -\mathrm{E}_{p(a,b)q(z_a \mid a,b)}\left[\log \frac{q(z_c \mid a, b)}{q(z_c \mid b)}\right]. \tag{18}$$

To stabilize optimization, an auxiliary distribution $r_b(z_s \mid y)$ is introduced. The auxiliary distribution $r_b(z_c \mid b)$ avoids direct optimization of complex or intractable target distributions. By introducing $r_b(z_c \mid b)$, an additional term is incorporated into the objective

$$\begin{aligned} &- I(a; z_c \mid b) \\ &= -\mathrm{E}_{p(a,b)\, q(z_c \mid a,b)}\left[\log \frac{q(z_c \mid a, b)}{r_b(z_c \mid b)} + \log \frac{r_b(z_c \mid b)}{q(z_c \mid b)}\right]. \end{aligned} \tag{19}$$

The auxiliary distribution reduces the gap between the target distribution and the inference distribution. In practice, the KL divergence between $q(z_c \mid a, b)$ and $r_b(z_c \mid b)$ penalizes inconsistency, which helps the model adjust its inference. At the same time, it enforces consistency between the two distributions and reduces bias during optimization.

The joint optimization of both regularization terms can then be expressed as

$$\begin{aligned} &(I(a; b; z_c) - I(z_a; z_c)) + (I(a; b; z_c) - I(z_b; z_c)) \\ &\geq \mathrm{E}_{p(a,b)}\left[\mathrm{E}_{q(z_c \mid a,b)\, q(z_a \mid a)}\left[\log p(a \mid z_a, z_c)\right]\right] \\ &\quad + \mathrm{E}_{p(a,b)}\left[\mathrm{E}_{q(z_c \mid a,b)\, q(z_b \mid b)}\left[\log p(b \mid z_b, z_c)\right]\right] \end{aligned}$$

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$$
\begin{aligned}
&- \mathrm{E}_{p(a,b)}\left[D_{\mathrm{KL}}(q(z_a \mid a) \,\|\, p(z_a))\right] \\
&- \mathrm{E}_{p(a,b)}\left[D_{\mathrm{KL}}(q(z_b \mid b) \,\|\, p(z_b))\right] \\
&- \mathrm{E}_{p(a,b)}\left[D_{\mathrm{KL}}(q(z_c \mid a,b) \,\|\, r_b(z_c \mid b))\right] \\
&- \mathrm{E}_{p(a,b)}\left[D_{\mathrm{KL}}(q(z_c \mid a,b) \,\|\, r_a(z_c \mid a))\right] \\
&+ H(A) + H(B). \qquad (20)
\end{aligned}
$$

The regularization terms are integrated into the ELBO objective. Reconstruction and disentanglement are optimized within a unified framework. Gradient updates are performed jointly.

Direct optimization of mutual information and KL terms is computationally expensive in high-dimensional latent spaces. The objective is reformulated into a joint maximization form similar to [38]. This formulation enables efficient variational inference. The resulting objective incorporates disentanglement regularization into the ELBO.

Reconstruction depends on the domain-specific representation $z_a$ and the shared representation $z_c$. The auxiliary modality $b$ does not participate in reconstruction. It provides supervision during training and improves latent representation quality.

The objective follows the rate–distortion principle. It consists of reconstruction error, coding rate for $z_a$, coding rate for $z_c$, and disentanglement regularization. The rate terms are reparameterized based on Shannon's theorem [42]. The final objective is expressed as

$$
\begin{aligned}
\mathcal{L} &= \mathrm{ELBO}(\theta,\phi) + \lambda\left(I(a;b;z_c) - I(z_a;z_c) - I(z_b;z_c)\right) \\
&= \lambda_1 \mathrm{E}\left[\log p(a \mid z_a, z_c)\right] - D_{\mathrm{KL}}(q(z_a \mid a) \,\|\, p(z_a)) \\
&\quad - D_{\mathrm{KL}}(q(z_c \mid a,b) \,\|\, p(z_c)) \\
&\quad + \lambda_2 D_{\mathrm{KL}}(q(z_c \mid a,b) \,\|\, r_a(z_c \mid a)) \\
&\quad + \lambda_2 D_{\mathrm{KL}}(q(z_c \mid a,b) \,\|\, r_b(z_c \mid b)) \qquad (21)
\end{aligned}
$$

where $\lambda_1$ balances rate and distortion, and $\lambda_2$ controls regularization strength. In experiments, $\lambda_2$ is fixed at 0.1.

Independent inference networks $r_a(z_c \mid a)$ and $r_b(z_c \mid b)$ predict the mean and variance of the shared latent variable $z_c$ from single-modality inputs. During training, the KL divergence between $q(z_c \mid a,b)$ and these auxiliary distributions is minimized. This constraint ensures that $z_c$ captures information consistent across modalities and suppresses redundancy.

The priors $p(z_a)$ and $p(z_c)$ serve as rate terms. KL penalties align posteriors with priors, controlling the entropy of latent variables and improving compression efficiency. Higher KL values indicate greater latent entropy, corresponding to more bits for storage. Minimizing KL divergence produces compact latent codes that retain essential information while supporting efficient representation learning.

## IV. Proposed Image Compression Method

### A. Overall Framework

The detailed architecture is shown in Fig. 5. The FE, PE, and CE are designed as alternating linear and nonlinear modules.

In the proposed framework, the analysis transform ($g_a$) projects HSI features into an independent space, generating the domain-specific latent representation $z_a$. Both HSI and LiDAR images are then processed by the FE, which produces the low-level shared features $F_a$ and $F_b$. Two PE transform these features into domain-specific higher level representations $z_{c_a}$ and $z_{c_b}$, capturing unique domain information. Meanwhile, $F_x$ and $F_y$ are also used to derive the shared high-level representation $z_c$, which reflects the common features across domains.

In the latent space, domain alignment regularization ($kld_loss$) is applied to align multimodal features, facilitating coherent and discriminative representation of shared semantics. The shared latent $z_c$ and the domain-specific latent $z_a$ are passed through the hyper encoder ($h_a$) and hyper decoder ($h_s$) for entropy modeling. The latent distributions are assumed to follow Gaussian priors. Thus, the parameterized priors $p(z_a)$ and $p(z_c)$ are estimated and incorporated as rate terms into the loss function.

The FE employs a selective nearest-neighbor matching mechanism, replacing traditional local attention. This mechanism avoids complex cross-modal self-attention and instead applies precise algorithmic matching between image blocks. It enables efficient extraction of shared features and effective handling of cross-modal relationships.

### B. Best Neighbor Matching Mechanism

In the domain of multimodal image processing, Transformer architectures [43] are extensively employed to facilitate cross-modal feature representation and interaction. The fundamental principle involves calculating weighted attention scores by integrating query vectors derived from the primary modality with key and value vectors obtained from auxiliary modalities. This process generates cross-modal attention representations that enable effective extraction of shared features. Although effective in multimodal representation, direct aggregation of all reference tokens in 3-D space significantly reduces computational efficiency.

In image compression tasks, another limitation emerges. Image blocks that are spatially distant across modalities often lack semantic correlation and exhibit very limited similarity. Relying on global self-attention under these conditions introduces two major drawbacks. First, the $O(N^2)$ complexity results in a heavy computational and memory burden. Second, redundant interactions arising from weakly correlated distant blocks attenuate attention weights and diminish the model's capacity to effectively capture essential local features.

To mitigate these challenges, we introduce an efficient block-matching mechanism designed for enhanced local feature extraction. The mechanism restricts attention connections to adjacent blocks. It enhances the modeling of local salient regions and strengthens their representation, as shown in Fig. 6.

The input feature map is partitioned into nonoverlapping blocks. For a block centered at $(x_c, y_c)$, its neighborhood features are collected within a square window of radius $r$, forming the candidate set

$$
\begin{aligned}
&A_{x,y} \\
&= \{F_i(x,y) \mid x_c - r \le x \le x_c + r,\ y_c - r \le y \le y_c + r\}. \qquad (22)
\end{aligned}
$$

where $A_{x,y}$ is the candidate region and $F_i$ denotes the feature map of the current block. The region is further divided into $N$

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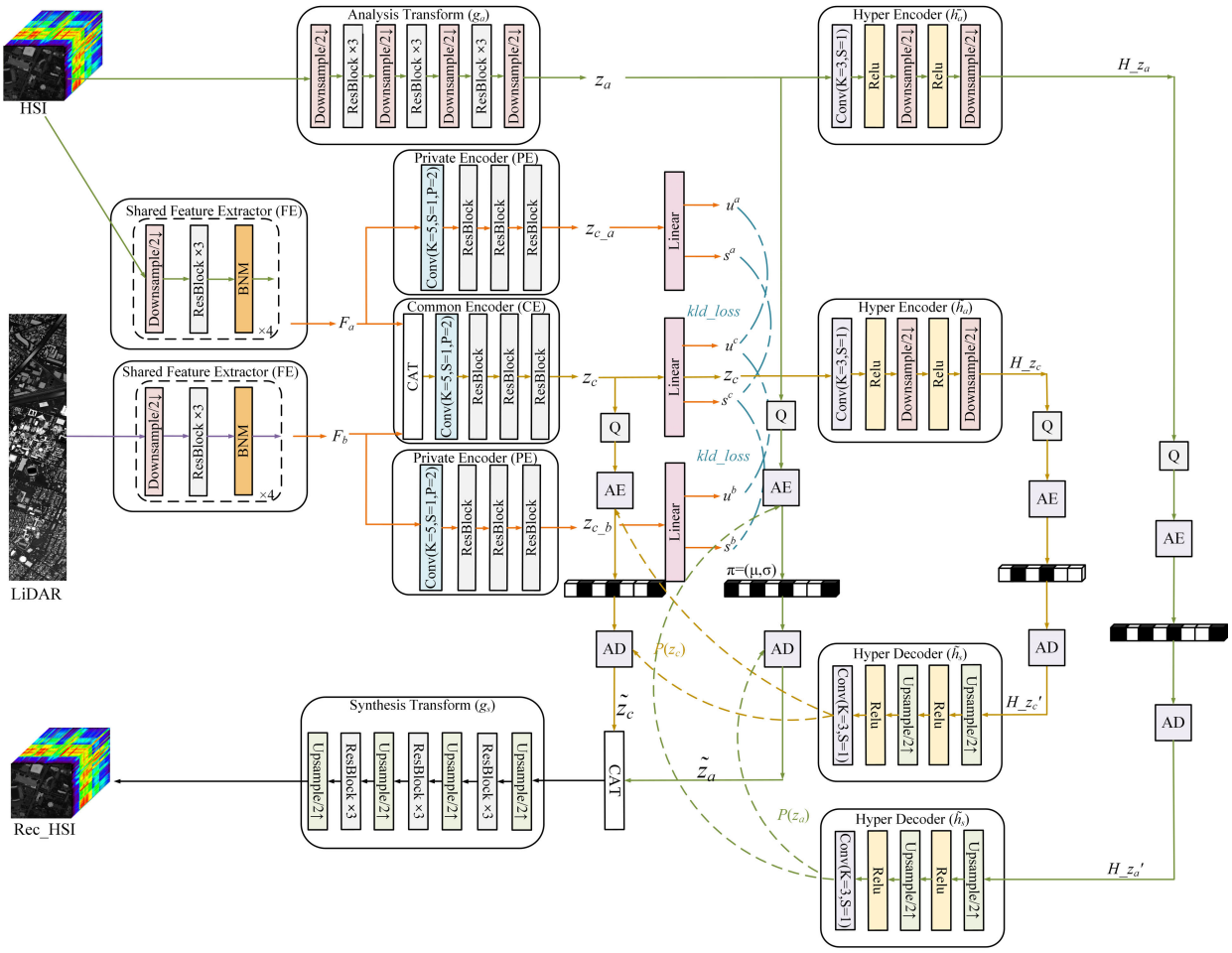



Fig. 5. Architecture of the proposed multimodal latent disentanglement compression model. CAT indicates channel concatenation. BNM refers to the best neighbor matching mechanism.




Fig. 6. Proposed best neighbor matching mechanism. Input feature maps are divided into blocks. For each block, neighborhood features are extracted, and candidate blocks are selected. The most similar blocks are concatenated with the current block, followed by a convolution layer to enhance feature representation.

candidate blocks

$$A_{x,y}^{i} = \left[A_{x,y}^{1}, A_{x,y}^{2}, \ldots, A_{x,y}^{N}\right]. \tag{23}$$

Similarity between the current block $F_i$ and each candidate block $A_{x,y}^{i}$ is measured by Euclidean distance

$$d\left(F_i, A_{x,y}^{i}\right) = \sqrt{\sum_{m=1}^{H}\sum_{n=1}^{W}\left(F_i(m,n) - A_{x,y}^{i}(m,n)\right)^2} \tag{24}$$

where $H$ and $W$ denote the height and width of the image block, respectively, and $m$ and $n$ represent the row and column indices within the block. Smaller distances indicate higher similarity.

The $K$ most similar candidate blocks are then selected, indexed by $i_1, i_2, \ldots, i_K$. These blocks are concatenated with $F_i$ along the channel dimension and further processed with a convolutional layer

$$F_i^{\text{new}} = \text{Conv}_{1\times1}\left[\text{Concat}\left(F_i, A_{x,y}^{i_1}, A_{x,y}^{i_2}, \ldots, A_{x,y}^{i_K}\right)\right]. \tag{25}$$

This mechanism avoids the computational burden of global self-attention. It focuses on local key regions, improving efficiency in cross-modal feature interaction. The final convolution layer enhances representation quality by extracting salient information from the aggregated blocks.

## V. Experimental Results and Analysis

The proposed model is evaluated on standard multimodal remote sensing datasets. Its performance is benchmarked against classical compression methods, including JPEG and BPG, as well as deep learning-based models such as Ballé et al. [44] (ICLR 2018), He et al. [45] (CVPR 2021), Cheng et al. [46] (CVPR 2021), and He et al. [47] (CVPR 2022). Additionally, the comparisons include recent state-of-the-art networks in the natural image compression domain, such as Jiang et al. [48] (ICML 2023) and Han et al. [49] (NeurIPS 2024), as well as the latest studies in hyperspectral image compression, including Zhang et al. [36] (GRSL 2024), Liu et al. [35] (JSTARS 2025), Liu et al. [9] (TGRS 2025),

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MSAHiFiC [50] (TGRS 2025), and SSCTG [13] (TCSVT 2026).

### A. Dataset

*1) Houston Dataset:* This dataset from the Houston area includes LiDAR-derived elevation data describing surface structures. The HSI ($349 \times 1905 \times 144$) contains 144 bands covering 380–1050 nm, capturing the spectral characteristics of urban materials.

*2) Trento Dataset:* This dataset from the Trento region includes LiDAR elevation data for detailed terrain modeling. The HSI ($166 \times 600 \times 63$) contains 63 bands covering 400–950 nm and spans about 100 ha with six land-cover classes.

It is noteworthy that hyperspectral data differ in both acquisition locations and sensor types. During preprocessing, invalid spectral bands, such as water vapor–affected regions, are typically removed, which leads to variations in the final number of bands across datasets. To ensure experimental consistency, all datasets are preprocessed in a uniform manner. HSIs are cropped to a size of $256 \times 256 \times 3$, with zero padding applied when necessary. The associated LiDAR images are adjusted to align with the spatial dimensions of the cropped hyperspectral data, producing a resolution of $256 \times 256$. A mixed validation strategy is adopted, and the training and test sets are strictly separated in spatial distribution to enhance the generalization and reliability of the experimental results.

### B. Model Setting and Evaluation Metrics

All learning-based methods are implemented using the PyTorch framework and trained on an NVIDIA GTX 4090 GPU with the Adam optimizer. Training consists of 5000 iterations, with eight mini-batches per iteration. The same learning rate schedule is applied to all deep learning-based methods to maintain fair comparison conditions.

In the proposed method, different $\lambda$ values (256, 2048, 8192, 16 384, and 45 000) are selected for the mixed dataset to accommodate various compression ratios. Network parameters are fixed as $N = 128$ and $M = 192$, where $N$ denotes the number of convolutional layers in the encoder and $M$ corresponds to the hyperprior module. For the methods proposed by Jiang et al. [48] (ICML 2023) and Han et al. [49] (NeurIPS 2024), the original configurations use $N = 192$ and $M = 320$. To ensure fair comparisons, all competing models are uniformly adjusted to $N = 128$ and $M = 192$, enabling performance evaluation under identical network capacities.

The compression performance of hyperspectral images is evaluated in terms of bits per pixel per band. Reconstruction quality is assessed using three standard metrics: peak signal-to-noise ratio (PSNR) [52], multi scale structural similarity index measure (MS-SSIM) [53], and spectral angle mapper (SAM) [54].

### C. Experimental Result

*1) Performance on Rate–Distortion and SAM:* The performance is evaluated under the same experimental settings, and the results are shown in Fig. 7. Due to the lack of standardized benchmarks for multimodal hyperspectral image compression, representative single-modality methods are included for reference comparison.

TABLE I

QUANTITATIVE PERFORMANCE COMPARISON OF BD-RATE (%), BD-PSNR (DB), BD-MS-SSIM (DB), AND BD-SAM ON THE MIX HSI DATASET (BPG [6] IS USED AS THE BASELINE)

| Method | BD-Rate (%) ↓ | BD-PSNR (dB) ↑ | BD-MS-SSIM (dB) ↑ | BD-SAM ↓ |
|---|---|---|---|---|
| Ours | -32.45 | 1.18 | 2.54 | -0.0031 |
| Liu et al. (TGRS-25) [9] | -18.72 | 0.62 | 0.98 | -0.0012 |
| Liu et al. (JSTARS-25) [36] | -14.10 | 0.45 | 0.82 | -0.0018 |
| Han et al. (NIPS-24) [50] | -10.55 | 0.32 | 0.55 | -0.0022 |
| Zhang et al. (GRSL-24) [37] | -8.24 | 0.28 | 0.41 | -0.0009 |
| Jiang et al. (ICML-23) [49] | -5.18 | 0.15 | 0.24 | -0.0011 |
| Cheng et al. (CVPR-21) [47] | -3.12 | 0.11 | 0.18 | -0.0007 |
| He et al. (CVPR-22) [48] | 12.35 | -0.42 | -1.15 | -0.0004 |
| He et al. (CVPR-21) [46] | 6.45 | -0.21 | -0.34 | 0.0005 |
| Ballé et al. (ICLR-18) [45] | 28.60 | -0.85 | -1.82 | 0.0025 |
| JPEG [4] | 145.20 | -4.12 | -6.50 | 0.0124 |

As shown in Fig. 7, the proposed method achieves superior overall rate–distortion performance across PSNR, MS-SSIM, and SAM. In terms of PSNR, it consistently outperforms existing methods over a wide range of bitrates, with comparable performance at low bitrates and more pronounced gains at higher bitrates. Under the MS-SSIM metric, the method maintains leading performance across all bitrates, indicating improved perceptual quality and structural consistency of the reconstructed images.

For the SAM metric, the method yields lower spectral angle errors across different bitrates, demonstrating effective preservation of spectral information while maintaining spatial structures. Overall, the results confirm the advantage of the proposed framework in achieving a favorable tradeoff between reconstruction fidelity, perceptual quality, and spectral consistency.

*2) Comparison of Visualization Results:* As shown in Figs. 8 and 9, traditional compression methods such as JPEG2000 [4] and BPG [6] often produce noticeable blocking artifacts and local blurring in compressed images, particularly in high-frequency and texture-rich regions. These artifacts indicate the inherent limitations of traditional codecs in balancing compression efficiency and visual fidelity, leading to a loss of fine details and spectral information.

Deep learning-based compression methods, including He et al. [45] and Cheng et al. [46], improve overall compression performance but remain less effective in reconstructing high-frequency details. They also require higher bit consumption, making it difficult to achieve an optimal balance between efficiency and quality. Although Jiang et al. [48] and Han et al. [49] preserve more local structures at comparable bitrates and demonstrate competitive performance, they consume more bits than the proposed model.

In contrast, the proposed multimodal disentanglement framework separates shared and private features and employs a lightweight entropy model for efficient latent representation coding. This design maximizes the preservation of image details and structural information under the same compression ratio.

To overcome the limitations of the RD curve in quantitative analysis, quantitative gain metrics such as BD-Rate, BD-PSNR, BD-MS-SSIM, and BD-SAM are introduced, which is shown in Table I. The BPG algorithm is used as the baseline to evaluate the overall rate-distortion performance of each model.

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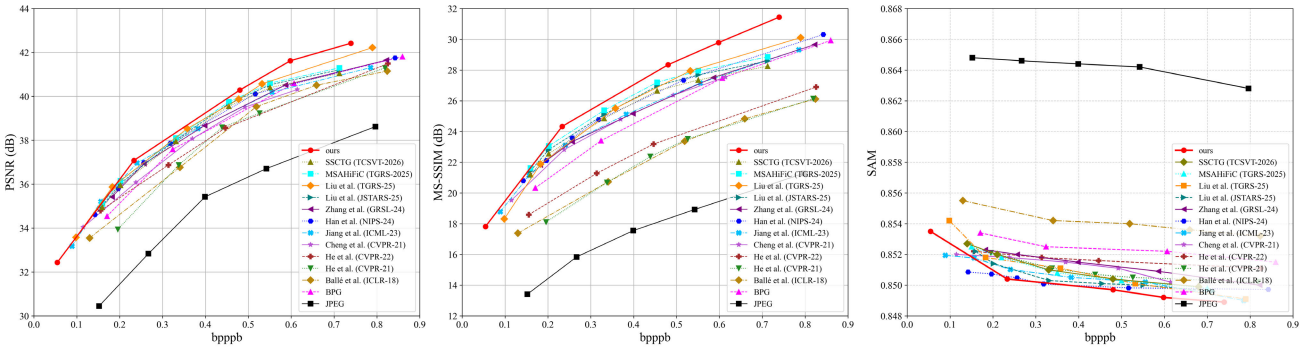



Fig. 7. Performance analysis of three standard metrics on the mixed dataset.

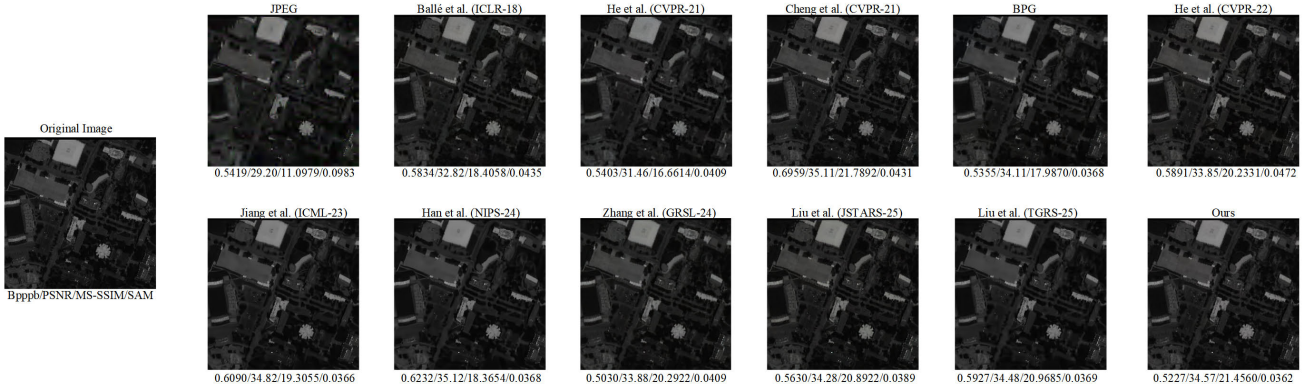



Fig. 8. Visual reconstruction results of the full 256 × 256 image region under high bitrate conditions.

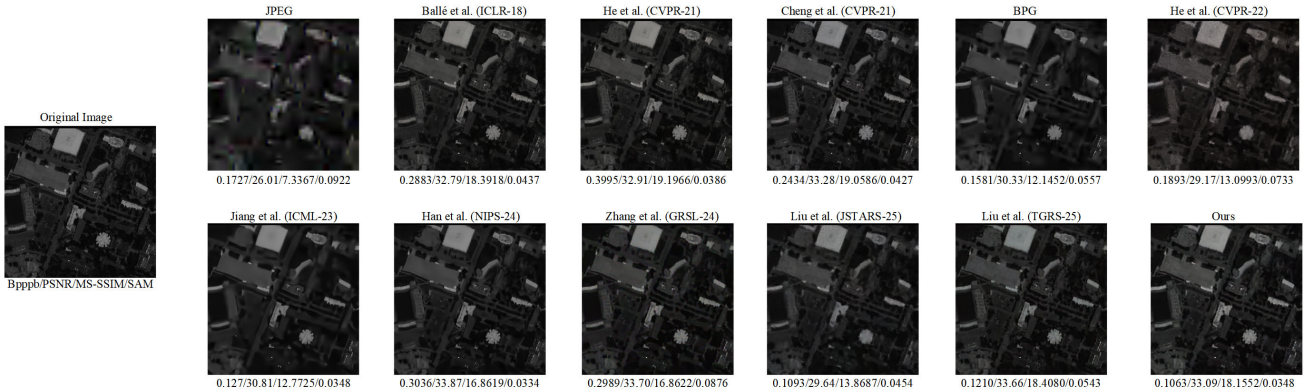



Fig. 9. Visual reconstruction results of the full 256 × 256 image region under low bitrate conditions.

*3) Model Complexity:* In this section, a comprehensive comparison of model complexity is conducted across different methods. The comparison includes the number of parameters (Parameters, M), computational complexity (FLOPs, G), encoding time (Encoding time, s), and decoding time (Decoding time, s) for different methods. The number of parameters and FLOPs reflect model scale and theoretical computational complexity. Encoding and decoding time characterize runtime efficiency from a practical perspective.

In addition, GPU memory consumption during training (In Training, GB) and inference (In Inference, MB) is reported. These metrics provide a more comprehensive evaluation of resource requirements in real-world deployment.

Table II reports that the proposed method has a larger parameter size than some lightweight models. The computational complexity remains moderate at 16.82 FLOPs, with encoding and decoding times of 0.021 and 0.010 s. These results demonstrate high runtime efficiency.

Compared with high-complexity models, such as those reported in [48] and [49], the proposed method achieves competitive compression performance with significantly reduced computational cost. GPU memory consumption remains within a controllable range during both training and inference. This result further demonstrates the practicality of the proposed method for real-world implementation.

*4) Results Under Different Representation Strategies:* To provide an intuitive analysis of the disentanglement strategy in latent-space modeling, Fig. 10 presents t-SNE visualizations of modality-specific features $z_a$ and cross-modal shared features $z_c$ for the hyperspectral modality. The proposed framework produces a clear separation between $z_a$ and $z_c$ in the embedding space. The corresponding samples form relatively indepen-

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TABLE II
COMPARATIVE ANALYSIS OF MODEL COMPLEXITY ACROSS MULTIPLE METRICS

| Method | Parameters (M) | FLOPs (G) | Enc-time (s) | Dec-time (s) | In Training (GB) | In Inference (MB) |
|---|---|---|---|---|---|---|
| JPEG [4] | - | 1.18 | 0.92 | - | - | - |
| BPG [6] | - | 1.09 | 0.75 | - | - | - |
| Ballé et al. (ICLR 2018) [45] | 5.08 | 12.34 | 0.248 | 0.023 | 7.74 | 844.36 |
| He et al. (CVPR 2021) [46] | 12.15 | 28.55 | 0.4124 | 0.0564 | 8.81 | 1361.43 |
| Cheng et al. (CVPR 2021) [47] | 9.70 | 27.49 | 0.314 | 0.025 | 9.03 | 1278.03 |
| He et al. (CVPR 2022) [48] | 33.79 | 54.65 | 0.351 | 0.035 | 9.51 | 1642.11 |
| Jiang et al. (ICML 2023) [49] | 118.41 | 82.36 | 0.428 | 0.156 | 17.30 | 1835.43 |
| Han et al. (NeurIPS 2024) [50] | 64.82 | 125.83 | 0.589 | 0.256 | 18.50 | 1832.01 |
| Zhang et al. (GRSL 2024) [37] | 20.52 | 28.96 | 0.3420 | 0.1280 | 13.85 | 1210.31 |
| Liu et al. (JSTARS 2025) [36] | 16.14 | 13.40 | 2.1130 | 0.2630 | 14.60 | 429.29 |
| Liu et al. (TGRS 2025) [9] | 21.89 | 26.82 | 0.059 | 0.030 | 20.63 | 1388.43 |
| Ours | 40.18 | 16.82 | 0.021 | 0.010 | 13.24 | 1152.52 |

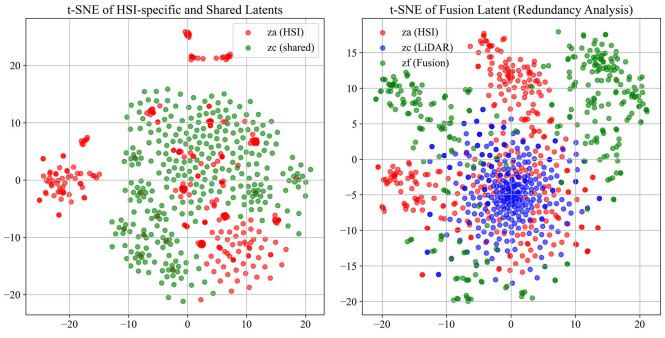



Fig. 10. Latent space distributions under disentanglement and fusion strategies.

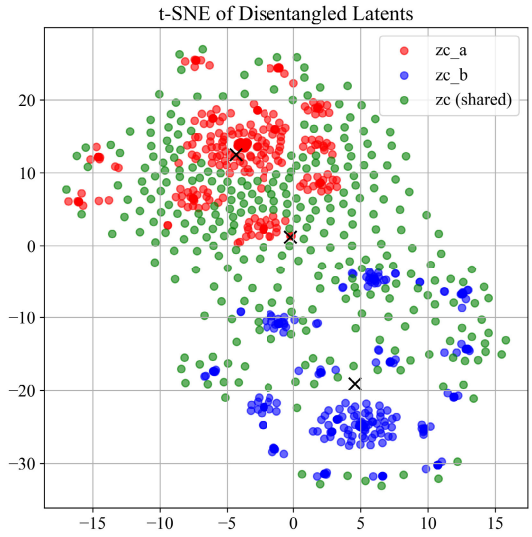



Fig. 11. t-SNE visualization of latent-space alignment and disentanglement.

dent cluster structures. This observation indicates effective discrimination between modality-specific information and cross-modal shared information in the latent space. The model enables explicit modeling of different information components. From a geometric perspective, this phenomenon reflects the structural effect induced by disentanglement constraints. It also reduces interference across different information components.

In contrast, the fusion strategy derives latent representations through direct mapping of multimodal features. This approach lacks explicit structural constraints. The t-SNE visualization shows significant overlap and mixed distributions among different information types. Clear cluster boundaries are difficult to identify. This distribution pattern indicates coupling between shared and modality-specific information in the latent space. The resulting representation exhibits limited discriminability.

The above analysis demonstrates effective separation between shared and modality-specific information through structured latent variable modeling. Representation-level disentanglement alone remains insufficient for stable utilization of shared information. Under single-modality input, accurate inference of shared latent variables becomes challenging. This limitation arises when the shared representation depends on joint distributions. Additional constraints on the inference process are required. The shared latent variable should maintain cross-modal consistency. It should also support approximate inference from single-modality input.

To address this issue, an optimization objective based on conditional mutual information is introduced, denoted as $-I(a; z_c \mid b)$. This objective enforces constraints on the conditional mutual information term. Two independent inference networks, $r_a(z_c \mid a)$ and $r_b(z_c \mid b)$, are constructed. These networks approximate the conditional distributions of the shared latent variable from single-modality inputs. Consistency between single-modality inference and joint inference is achieved by minimizing the following KL divergence terms

$$D_{KL}\left(q\left(z_c \mid a, b\right) \| r_a\left(z_c \mid a\right)\right) + D_{KL}\left(q\left(z_c \mid a, b\right) \| r_b\left(z_c \mid b\right)\right). \tag{26}$$

The core function of this constraint lies in suppressing the dependence of the shared latent variable $z_c$ on modality-specific information from a single modality. This effect is enforced under the condition of the other modality. The shared latent variable is thus encouraged to capture information that remains stable across modalities. As a result, $z_c$ produces consistent inference outcomes under single-modality input. This design avoids reliance on information recoverable only from joint modalities. It further promotes effective modeling of shared information and suppresses redundant components.

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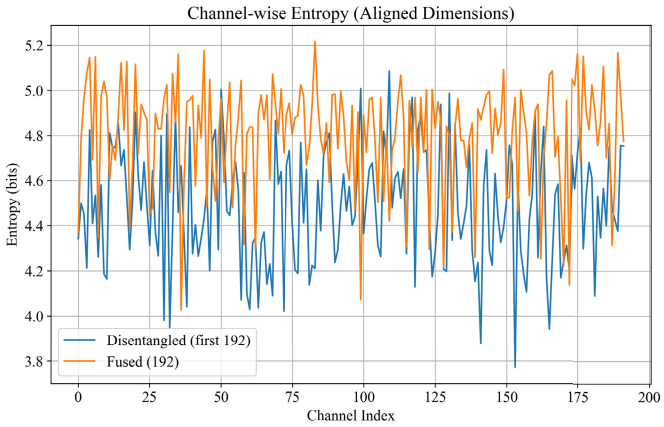



Fig. 12. Channel-wise entropy distribution under different strategies.

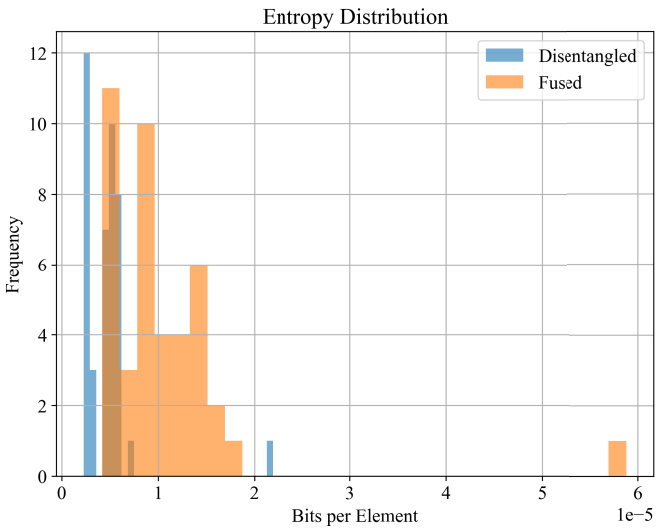



Fig. 13. BPE distribution under different strategies.

To provide an intuitive illustration of this mechanism in the latent space, t-SNE visualizations of $z_{c_a}$, $z_{c_b}$, and $z_c$ are presented in Fig. 11. Here, $z_{c_a}$ and $z_{c_b}$ denote shared latent variables inferred from modalities A and B. The variable $z_c$ denotes the shared representation inferred from joint modalities.

The visualization shows strong neighborhood proximity and consistent distribution patterns between $z_{c_a}$ and $z_c$ in the feature space. This observation agrees with the imposed Kullback–Leibler divergence constraint. The single-modality inference distribution is statistically aligned with the joint distribution. This alignment appears as similarity in geometric locations and local structures in the embedding space. The result provides qualitative evidence that the shared latent variable can stably represent cross-modal consistent structural features using only primary modality A.

In contrast, $z_{c_b}$ maintains a clear separation from $z_c$ in the global distribution. The cluster centers show noticeable offsets. This pattern indicates limited encoding of auxiliary modality B information into the shared latent space. The shared representation avoids interference from modality-specific information. This phenomenon reflects the role of the proposed mutual information constraint. Redundant information propagation across latent variables is suppressed. The disentangled structure in the latent space is thus preserved.

These observations are consistent with the design objective of the proposed method. The HSI modality serves as the primary source for constructing the shared representation. The LiDAR modality functions as an auxiliary constraint during training. It guides the learning of shared features without directly dominating the compressed representation. This design improves interpretability and structural consistency of latent representations while preserving expressive capacity.

To provide an intuitive comparison of different multimodal fusion strategies, latent representations from the fusion strategy and the disentanglement strategy are extracted. A channel-wise quantitative analysis of information distribution is then conducted. The entropy of each channel is computed to characterize information content and activation levels. The results are visualized to illustrate channel-wise information distributions under different strategies. Fig. 12 presents the channel entropy statistics for both approaches.

The visualization reveals pronounced entropy peaks in several channels under the fusion strategy. These peaks indicate repeated encoding of information within specific channels. The latent representation contains redundancy. This observation suggests that simple multimodal fusion mixes shared and modality-specific information within the same channels. Such mixing reduces representation efficiency in the latent space.

In contrast, the disentanglement strategy produces a more balanced and smoother entropy distribution. The number of high-entropy channels is significantly reduced. This pattern indicates effective separation between cross-modal shared information and modality-specific information. Redundant encoding is reduced, and information utilization is improved.

To further quantify the information efficiency of latent representations, the bit-per-element (BPE) distribution is computed for both strategies. The results are visualized in Fig. 13.

The disentanglement strategy exhibits a more compact and concentrated BPE distribution. The average BPE is lower than that of the fusion strategy. This result indicates that the disentangled latent space encodes essential information with fewer bits while maintaining representation capability. This finding is consistent with the channel entropy analysis. It further demonstrates that the disentanglement strategy improves multimodal compression efficiency while preserving information completeness and precision. In summary, the disentanglement strategy outperforms the fusion strategy in redundancy suppression and compression performance. It provides a more effective latent-space modeling approach for multimodal image compression.

To analyze the difference in information preservation between compression strategies, a comparative study is conducted under an approximately equal bitrate condition (0.3 bpppb). A pixel located at spatial position (128, 128) from the Trento dataset is selected. Its reconstructed spectral curves are visualized. Spectral continuity is further evaluated across different methods. The results are presented in Fig. 14.

The disentanglement method produces reconstructed spectra that closely match the ground-truth curve across all bands. The variation trend remains smooth and continuous. No obvious

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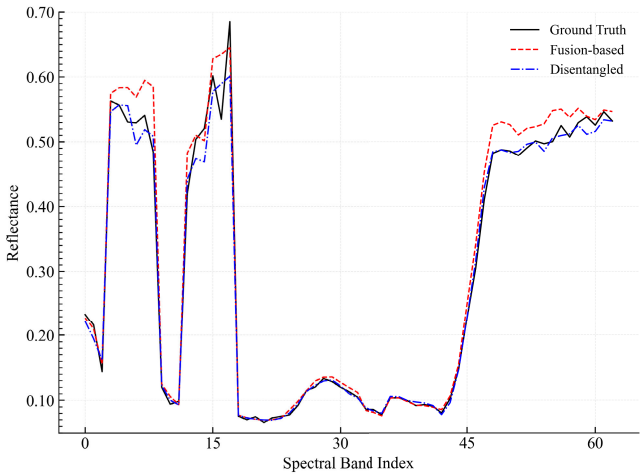



Fig. 14. Spectral consistency comparison.

oscillation or distortion is observed. The fusion strategy shows larger deviations and reduced smoothness.

These results indicate that the disentanglement mechanism effectively separates cross-modal shared information and modality-specific information. Information mixing and redundant interference are reduced during compression. The method preserves spectral details while avoiding excessive smoothing. High spectral consistency is maintained under high compression ratios.

*5) Ablation Study on Network Architecture:* To further analyze the contribution of each component, rate–distortion performance under different model configurations is evaluated. As illustrated in Fig. 15, the results are compared across several representative settings.

1) Baseline denotes the multimodal disentangled hyperspectral image compression model without the best neighbor matching mechanism.
2) Baseline + Best Neighbor Matching denotes the proposed disentanglement method with multimodal alignment.
3) Fusion denotes the multimodal compression approach based on feature fusion.
4) Baseline-w/o Constraint denotes the model without the information-theoretic constraint.

The introduction of the best neighbor matching mechanism (Baseline + Best Neighbor Matching) leads to the best performance across all bitrates. The improvement becomes more significant in the medium-to-high bitrate range. The method consistently outperforms the Baseline in terms of PSNR, which indicates reduced pixel-level reconstruction error. A stable advantage is also observed in MS-SSIM, which reflects improved structural preservation and texture reconstruction. The SAM metric shows further reduction, which indicates enhanced spectral consistency. These results demonstrate that explicit cross-modal matching strengthens feature alignment in the latent space. The representation of shared information becomes more accurate, and redundant encoding is reduced.

The Baseline model maintains stable performance without the matching mechanism. This result indicates that the disentanglement framework itself provides strong representation capability. A certain degree of separation between shared and modality-specific information is already achieved. The Baseline-w/o Constraint model outperforms the Fusion method. This result indicates that disentanglement-based modeling provides more effective information representation than direct feature fusion. The observation highlights the importance of information-theoretic constraints in regulating latent-space distributions and suppressing redundancy. The model without constraints exhibits greater flexibility in representation. Its performance remains inferior to the full model. This comparison validates the effectiveness of the proposed constraint mechanism.

In contrast, the Fusion method shows weaker overall performance. Simple feature fusion fails to distinguish shared information from modality-specific information. Redundant and interfering components are introduced. Compression efficiency is reduced, and reconstruction quality is degraded. In addition, visual comparisons of reconstruction results for the four ablation models are provided. The results are shown in Fig. 16.

Fig. 16 presents reconstructed images under different model configurations. The Baseline + Best Neighbor Matching model achieves the best visual quality. It preserves structural details, recovers textures, and maintains cross-modal consistency. The reconstructed scenes closely match the original data. The Baseline model ranks second in performance. The Baseline-w/o Constraint model shows noticeable information degradation in certain regions. In contrast, the Fusion method produces reconstructions with blurred details and mixed information. Visual quality is relatively lower.

Overall, among the four configurations, Baseline + Best Neighbor Matching achieves the best performance in both quantitative metrics and visual quality. The Fusion strategy shows the weakest performance. Fusion-based methods demonstrate effectiveness in other multimodal tasks. Experimental results indicate limited suitability for multimodal hyperspectral image compression. This task requires strong spectral consistency and effective disentanglement of information components.

The ablation results demonstrate that each key module contributes to performance improvement. The combination of best neighbor matching and information-theoretic constraints leads to consistent gains across evaluation metrics. These components enhance cross-modal feature alignment and regulate latent-space distributions. Their effects are complementary. The overall compression performance is improved.

The attention heatmaps are generated from hyperspectral and LiDAR images using the best neighbor matching module and attention-based feature compression. They clearly capture the most informative regions in the images and provide an intuitive visualization of the model's attention distribution, as shown in Fig. 17.

The best neighbor matching computes dense similarities among local patches to identify critical regions and focuses attention on the most relevant features. The attention-based feature compression module compresses the attention features into a single-channel map. Convolutional layers are used for feature compression. The resulting map highlights regions with high attention responses.

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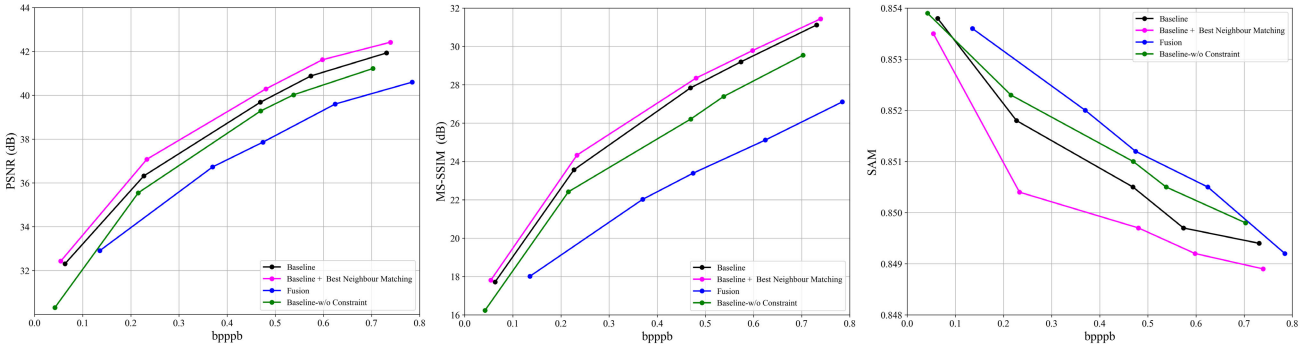



Fig. 15. Performance comparison in the ablation study.

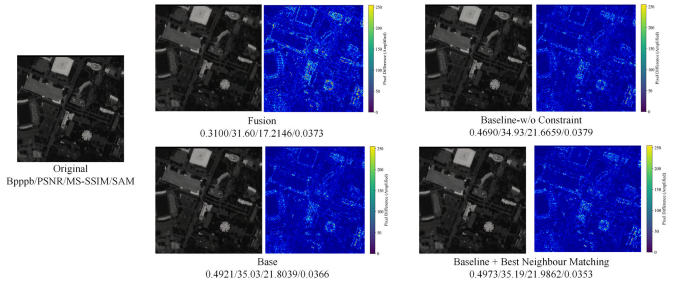



Fig. 16. Visual comparison of reconstruction results under four ablation settings.

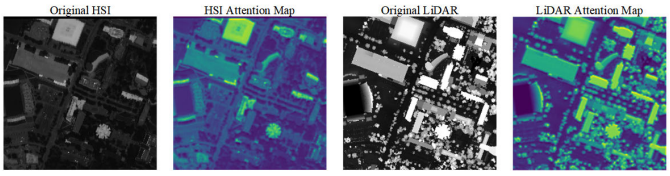



Fig. 17. Attention heatmaps for HSI and LiDAR images extracted using the best neighbor matching.

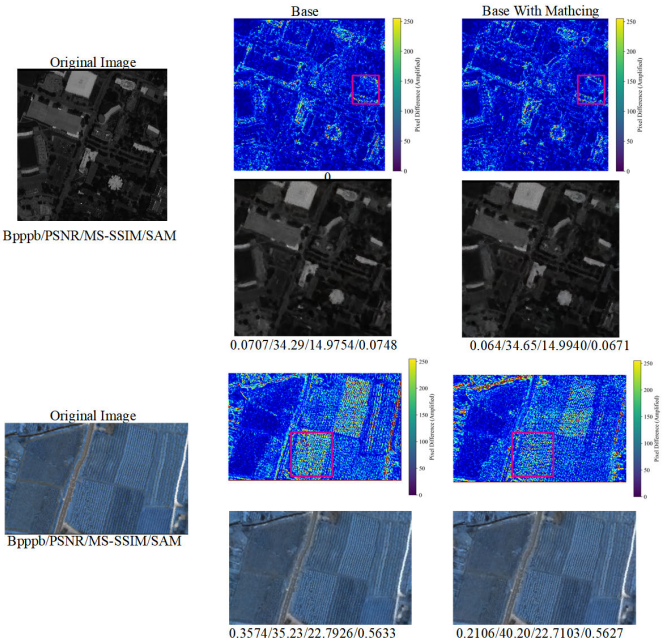



Fig. 18. Reconstructed results for ablation analysis. Top: reconstructed results of a $256 \times 256 \times 3$ Houston patch at low bit rates. Bottom: reconstructed results of a $256 \times 165 \times 3$ Trento patch at middle bit rates.

In the heatmaps, low-attention regions are shown in blue–violet tones, while high-attention areas appear in yellow–green. These visualizations effectively reveal the spatial distribution of attention. They show how the model emphasizes different regions in hyperspectral and LiDAR images. They also highlight the crucial role of the best neighbor matching module in guiding feature selection and improving multimodal feature representation.

Fig. 18 presents the pixel-level error distribution. They present reconstruction results for selected regions from the mixed dataset at extremely low bitrates. Error maps visualize the absolute pixel-wise differences between the original and reconstructed images. They show that errors concentrate along edges and high-frequency texture regions, such as roof boundaries and other complex structures. Smooth or low-frequency areas exhibit relatively small errors.

The highlighted regions mark areas with strong edge variations. A visual comparison shows that the proposed scanning mechanism captures and preserves local salient features more effectively. It achieves superior reconstruction quality in critical regions.

Compared with the ablated variants, edge regions exhibit fewer discontinuities and reduced error accumulation. Local structures are therefore better preserved. These results demonstrate that best neighbor matching guides the reconstruction toward spatially consistent neighborhoods. Misalignment and blurring at object boundaries are effectively suppressed.

*6) Cross-Dataset Generalization and Validation:* Two multimodal HSI–SAR datasets, Berlin [51] and Augsburg [51], are used for cross-dataset evaluation. The Berlin dataset contains HSI of size $797 \times 220$ pixels with 244 spectral bands covering 400–2500 nm and SAR data from Sentinel-1 dual-polarization (VV–VH). The Augsburg dataset contains HSI of size $332 \times 485$ pixels with 180 spectral bands from 400 to 2500 nm at 10-nm intervals, and SAR data from Sentinel-1 dual-polarization (VV–VH).

The trained HSI–LiDAR model expects a single-channel auxiliary input, while SAR data contain four polarization channels. The VV channel is used as the auxiliary input because it provides the strongest backscatter signal and is widely applied in remote sensing. Only this channel is employed to reduce modality dimensional mismatch. Model weights trained on LiDAR–HSI datasets are applied to the HSI–SAR datasets to evaluate cross-dataset generalization.

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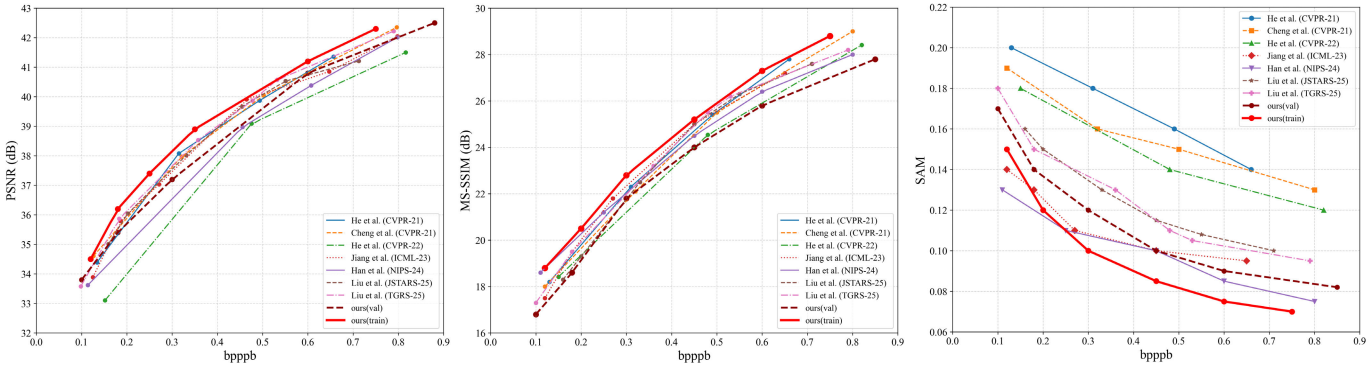



Fig. 19. Performance on two HSI–SAR datasets.

The proposed method maintains stable performance across all bitrate ranges in PSNR, MS-SSIM, and SAM. Performance is slightly lower than when using all SAR channels, but the results remain comparable to existing methods, with more pronounced advantages at medium-to-high bitrates (bpppb > 0.3), as shown in Fig. 19.

Cross-dataset validation, denoted as ours(val), evaluates the model on datasets not used during training. Performance shows a moderate decline relative to ours(train), reflecting differences in scene distribution, sensor response, and noise characteristics, which represent typical domain shifts. Despite this decline, ours(val) remains superior or comparable to existing methods across most bitrate ranges. SAM values remain low, indicating effective preservation of spectral structures. These results demonstrate stable cross-modal representation under cross-dataset conditions.

## VI. Discussion

Traditional single-modality hyperspectral image compression methods rely primarily on the statistical properties of spectral data. These methods do not exploit complementary information from additional modalities. This limitation leads to inherent performance constraints. One major issue is latent ambiguity. Different latent configurations may yield similar reconstructions but fail to preserve the true spatial–spectral structure. This effect becomes more severe at high bit rates. It results in fabricated textures, geometric distortions, and spectral deviations. Reconstruction fidelity degrades. The latent space loses physical consistency. In addition, single-modality frameworks cannot adequately capture complex cross-band correlations. This limitation restricts the modeling of structural and semantic relationships in hyperspectral data.

To overcome these limitations, we introduce a latent-space disentangled compression model under a maximum marginal likelihood formulation. LiDAR is incorporated as an auxiliary modality. It provides structural guidance for latent representations and promotes physically consistent and interpretable features. The framework employs modality-specific encoders and a shared encoder. These components decouple independent information streams. A mutual information-based regularization strategy suppresses redundancy. The optimization objective is derived from variational inference. This design ensures theoretical consistency and interpretable disentanglement. A dynamic nearest-neighbor matching mechanism further aligns shared structures across modalities. The alignment is based on Euclidean proximity. This mechanism improves the reliability and consistency of cross-modal representations.

Despite these advantages, several limitations remain. The performance depends on the quality of auxiliary modalities. Noisy or misaligned LiDAR data may degrade disentanglement and reconstruction accuracy. The current formulation relies on fixed information-theoretic constraints. This design limits flexibility under varying multimodal conditions. Computational and memory costs remain considerable. The issue becomes more significant for high-resolution hyperspectral data. This factor poses challenges for large-scale and real-time deployment.

Beyond compression performance, the proposed framework raises several broader considerations. These aspects include applicability to diverse hyperspectral tasks, robustness under practical conditions, compatibility with emerging foundation models, and connections to unified theoretical formulations.

From a task perspective, the proposed information-theoretic latent disentanglement framework exhibits strong generality. The method imposes structural constraints on latent representations. It decomposes the latent space into shared representations and modality-specific representations. This design enables structured information modeling. In hyperspectral image classification, the shared representation captures cross-modal consistent semantic information, such as class-related spectral features. The modality-specific representation preserves discriminative information unique to each modality. This disentanglement mechanism reduces redundant interference and improves class separability. In anomaly detection, the structured latent representation separates normal patterns and abnormal deviations. Shared components represent common structures. Modality-specific components capture residual deviations. This property improves anomaly sensitivity and detection accuracy. The framework also supports adaptive weighting between shared and private representations. This flexibility enables adjustment to different task requirements. Generalization ability and interpretability are improved in multitask scenarios.

Practical remote sensing applications involve nonideal conditions. Hyperspectral data often suffer from spectral variability, sensor noise, and complex imaging environments. These factors introduce instability in latent representations. Compression performance and downstream tasks may be

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affected. Future work will explore robust modeling strategies under such conditions. One direction involves adaptive information constraints. Constraint strength can vary according to the signal-to-noise ratio of spectral bands or modalities. Another direction involves robust regularization mechanisms. Noise-sensitive terms or distribution-shift penalties can be incorporated into the loss function. These designs improve stability under noise interference, sensor drift, and environmental variation. The latent representation remains structured and discriminative. Generalization performance improves in complex remote sensing scenarios. Compression efficiency and information fidelity are maintained.

Foundation models have shown rapid progress in remote sensing and hyperspectral imaging. Large-scale pretraining enables rich spatial–spectral feature extraction and high-order statistical modeling. These models provide strong prior representations for hyperspectral data. Integration with the proposed information-theoretic disentanglement framework represents a promising direction. The foundation model can serve as an encoder or FE. The latent representation then combines a disentangled structure with pretrained semantic and spectral knowledge. Representation capacity and transferability are improved. The shared latent space also supports seamless transfer to downstream tasks, including classification, object detection, and anomaly detection. Compression performance and cross-task capability can be jointly enhanced. Furthermore, prior knowledge from foundation models can be combined with information-theoretic constraints. Adaptive compression strategies can be designed. Bit allocation can adjust according to image content complexity and task requirements. Higher efficiency in multimodal compression can be achieved.

A unified theoretical framework remains an important direction. Multimodal representation learning, information-theoretic constraints, and rate–distortion optimization can be further integrated. Such a framework supports multitask and multimodal hyperspectral modeling. This direction improves compression performance. It also promotes efficient representation and understanding of multimodal remote sensing data.

## VII. Conclusion

This study proposes a latent-space disentangled compression model for multimodal hyperspectral images. LiDAR is incorporated as an auxiliary modality to guide latent representation learning. Mutual information-based regularization and variational inference optimization explicitly structure the latent space, ensuring physical consistency and interpretability. A dynamic nearest-neighbor matching mechanism aligns cross-modal features, enhancing reconstruction fidelity.

Extensive experiments on multiple multimodal datasets demonstrate that the proposed method achieves superior rate–distortion performance compared with existing single-modal and multimodal compression approaches. The model reduces redundant encoding, preserves spatial–spectral structures, and maintains stability under cross-dataset and cross-modal evaluations. The results establish a theoretically grounded and practically effective framework for multimodal hyperspectral image compression with disentangled latent representations.

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